Hierarchical Linguistic Knowledge in BERT
A Layer-wise Probing Analysis

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Abstract

We investigate the distribution of linguistic knowledge across the layers of a frozen BERT, a pre-trained transformer model, using layer-wise probing classifiers on three tasks: part-of-speech tagging (morphological), dependency relation labeling (syntactic), and sentiment analysis (semantic). By training classifiers on representations from each layer and employing scalar mixing weights and cumulative scoring, we map how different types of linguistic information are encoded throughout the network. Our results reveal a clear hierarchical organization: morphological information is concentrated in early layers, syntactic structure in middle layers, and semantic knowledge in upper layers. This stratification emerges naturally from pre-training without explicit supervision.

1 Introduction

Pre-trained transformer models like BERT [2] have revolutionized natural language processing, achieving state-of-the-art results across diverse tasks. However, despite their empirical success, the internal mechanisms and representations learned by these models remain poorly understood. Understanding what linguistic knowledge is encoded in different layers is crucial for both interpretability and improving model design.

Previous work has explored BERT's representations through probing classifiers [3,6], small supervised models trained to predict linguistic properties from frozen representations. These studies suggest that BERT layers capture different linguistic abstractions, but the exact nature of this hierarchy remains an active research question.

In this work, we conduct a systematic layer-wise probing analysis using three tasks that span the linguistic hierarchy: part-of-speech tagging (morphological), dependency relation labeling (syntactic), and sentiment analysis (semantic). We train logistic regression classifiers on representations from each of BERT's 12 transformer layers plus the embedding layer, but also use mixing scalar weights method proposed in to analyze each layer's contribution. This allows us to map how linguistic knowledge is distributed across layers.

Our main contributions are:

- A systematic analysis of linguistic knowledge across all BERT layers using three complementary probing tasks
- Empirical evidence that BERT learns a hierarchical linguistic pipeline: morphology (early layers) → syntax (middle layers) → semantics (late layers)
- Demonstration that this hierarchy emerges without explicit supervision during pre-training (BERT being frozen)

2 Related Work

The rapid adoption of pre-trained language models has spurred a sub-field of research often termed "BERTology," focused on interpreting the internal representations of Transformer models [5]. Our work builds primarily on methodologies developed for analyzing Contextual Word Representations (CWRs).

2.1 Probing Contextual Embeddings

Before the dominance of Transformer models, Peters et al. [4] introduced ELMo and demonstrated that deep bi-directional LSTMs learn a hierarchical representation of language. They found that lower layers of the network were specialized in local syntactic relationships (e.g., part-of-speech tagging), while higher layers better captured context-dependent semantic information (e.g., word sense disambiguation). This pivotal work established the layer-wise probing paradigm: training simple, diagnostic classifiers on top of frozen hidden states to measure the mutual information between the representation and a specific linguistic property.

2.2 Linguistic Hierarchy in BERT

Following the release of BERT [2], several studies applied similar probing techniques to Transformer architectures. Jawahar et al. [3] investigated the structure of BERT layers using a range of syntactic and semantic tasks. Their findings mirrored those of ELMo, suggesting that BERT captures phrase-level information in the lower-to-middle layers and linguistic structure in the middle layers, with semantic features appearing at the top (using Tensor Product Decomposition Networks to analyze the representations).

Most relevant to our study is the work of Tenney et al. [6], specifically "BERT Rediscovers the Classical NLP Pipeline." By employing *edge probing* [7] —a span-based probing framework—they quantified the amount of information available at each layer for various tasks. They introduced the use of scalar mixing weights (learning a weighted sum of layers) to identify the "center of gravity" for different linguistic tasks. Their results provided strong empirical evidence that deep language models relearn the traditional NLP pipeline: processing morphology first, followed by syntax, and finally high-level semantics and coreference resolution.

While Tenney et al. utilized complex span-based edge probing to handle tasks like Semantic Role Labeling (SRL), our work focuses on replicating these hierarchical findings using token-level and sentence-level logistic regression probes. We aim to verify if the same clear stratification of linguistic knowledge—Morphology $\rightarrow$ Syntax $\rightarrow$ Semantics—holds when analyzing standard single-token classification and pooling strategies.

It's also worth noting the work of Clark et al. [1] who analyzed BERT's attention mechanisms to understand what linguistic phenomena the model attends to at different layers. Their findings complement probing studies by showing that attention heads in lower layers focus on local syntactic relations, while higher layers capture longer-range dependencies and semantic relationships.

3 Methodology

To investigate the localization of linguistic knowledge within BERT, we adopt a layer-wise probing methodology. We freeze the parameters of the pre-trained encoder and train lightweight diagnostic classifiers on top of the representations extracted from each layer. This ensures that we are measuring the information intrinsic to the pre-trained model rather than the capacity of the probe to learn the task from scratch.

3.1 Probing Tasks

We select three tasks to cover the spectrum of linguistic complexity, from local morphology to global semantics:

Part-of-Speech (POS) Tagging. A morphological task where the goal is to assign a grammatical category (e.g., Noun, Verb, Adjective) to each token in a sentence. This task relies primarily on local word context and surface-level features.

Dependency Relation Labeling. A syntactic task that involves classifying the functional relationship (e.g., *nsubj*, *dobj*) between a head word and its dependent. We simplify this to a labeling task where the model must predict the relation type given the token representation, assessing the model's encoding of syntactic structure.

Sentiment Analysis. A semantic task requiring the classification of a whole sentence as positive or negative. Unlike the previous two tasks, this requires aggregating information across the entire sequence to form a high-level representation of abstract meaning.

3.2 Feature Extraction

We use the pre-trained `bert-base-uncased` model (12 layers, 768 hidden size). For a given input sequence of length T, the model produces a set of hidden states $H^{(l)} = [h_1^{(l)}, \ldots, h_T^{(l)}]$ for each layer $l \in \{0, \ldots, 12\}$, where layer 0 corresponds to the non-contextual static embeddings.

For token-level tasks (POS and Dependency), we extract the hidden state corresponding to the target token. For the sentence-level task (Sentiment), we extract the representation of the special `[CLS]` token, which is designed to capture aggregate sequence information.

3.3 Probing Architectures

Layer-wise Logistic Regression. To establish a baseline of information availability, we train a simple Logistic Regression classifier independently for each layer $l \in \{0, \ldots, L\}$. For a target representation $h^{(l)}$, the probability of class c is estimated via:

$$P(y = c|h^{(l)}) = \text{softmax}(W^{(l)}h^{(l)} + b^{(l)}) \tag{1}$$

where $W^{(l)}$ and $b^{(l)}$ are layer-specific parameters. This approach allows us to trace the "performance trajectory" across the network, identifying specific layers where the representation becomes most linearly separable for a given task, independent of other layers.

Scalar Mixing Weights. To quantify the relative importance of each layer more rigorously, we adopt the scalar mixing technique proposed by Tenney et al. [6]. Instead of probing layers individually, we train a classifier on a weighted sum of all layers:

$$h_{mix} = \gamma \sum_{l=0}^{L} s_l h^{(l)} \tag{2}$$

where γ is a trainable scaling parameter and s_l are normalized weights computed via a softmax over learnable parameters α_l: $s_l = \frac{\exp(\alpha_l)}{\sum_k \exp(\alpha_k)}$. The mixing weights s_l are learned jointly with the classifier. By analyzing the magnitude of s_l after training, we can determine which layers the model "prefers" for solving a specific task.

Cumulative Scoring. While mixing weights reveal which layers the model attends to, they do not necessarily indicate where information is first generated. Information about a specific linguistic phenomenon (e.g., syntax) may be spread across multiple layers or refined gradually. To disentangle discovery from utilization, we employ *Cumulative Scoring*.

We train a series of L separate classifiers, where the l-th classifier has access to the scalar mix of all layers up to l (i.e., layers $0, \ldots, l$). This allows us to observe the *differential gain* or "delta" ($\Delta^{(l)}$) in performance provided by adding layer l:

$$\Delta^{(l)} = \text{Score}_l - \text{Score}_{l-1} \quad (3)$$

where Score_l is the metric (e.g., Accuracy) of the cumulative classifier at layer l. A high $\Delta^{(l)}$ indicates that layer l contributes novel information not present in the preceding layers.

3.4 Comparison with Edge Probing

Our approach draws inspiration from the *Edge Probing* framework introduced by Tenney et al. [7]. However, unlike Edge Probing, which requires complex span pooling and projection layers to handle multi-token constituents, we restrict our analysis to single-token and `[CLS]` token classification. This simplification allows for a more direct comparison of the raw distributional properties of BERT's hidden states without the potential confound of complex pooling operations.

3.5 Metrics

We evaluate task performance using Accuracy. To summarize the layer-wise distribution of information, we compute three summary statistics:

Center of Gravity (CoG). We compute the CoG for the mixing weights s to calculate the average layer attended to by the full model:

$$\bar{E}_s[l] = \sum_{l=0}^{L} l \cdot s_l \quad (4)$$

Expected Layer. To determine where information is typically resolved, we compute the Expected Layer over the differential scores (Δ). As observed by Tenney et al., Layer 0 (embeddings) often provides a massive performance jump that can skew the expectation. Therefore, we calculate the expectation over the contextual layers ($l \geq 1$):

$$\bar{E}_\Delta[l] = \frac{\sum_{l=1}^{L} l \cdot \Delta^{(l)}}{\sum_{l=1}^{L} \Delta^{(l)}} \quad (5)$$

A gap between the CoG and the Expected Layer suggests a difference between where information is *discovered* (Expected Layer) and where it is *refined* for use (CoG).

KL Divergence. Finally, to measure how "localized" the information is, we compute the Kullback-Leibler (KL) divergence between the learned distribution (either weights s or normalized deltas Δ) and a uniform distribution U:

$$KL(P||U) = \sum_l P(l) \log \frac{P(l)}{U(l)} \quad (6)$$

A high KL divergence indicates that the linguistic knowledge is concentrated in a few specific layers, while a low KL divergence suggests the information is diffused throughout the network.

4 Results

4.1 Layer-wise Performance Trajectories

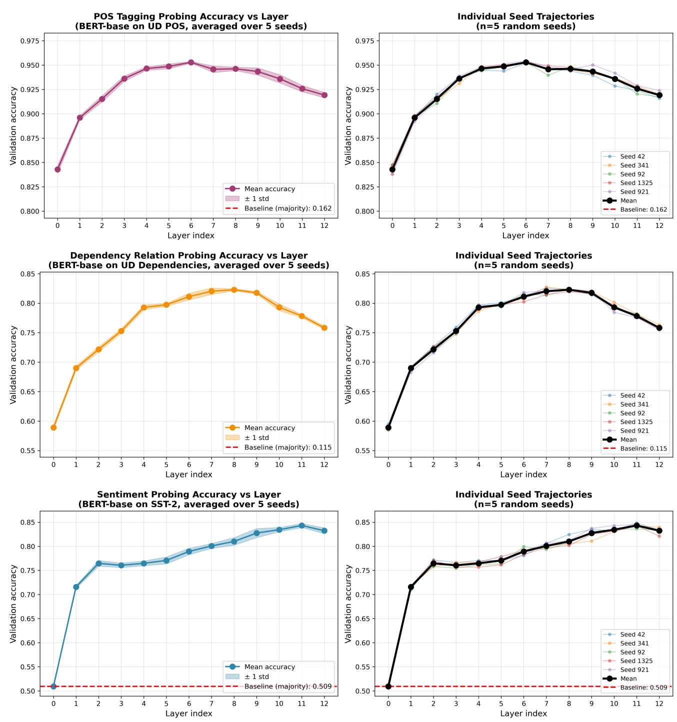

Figure 1: Layer-wise probing performance for POS tagging (top), Dependency Relation Labeling (middle) and Sentiment Analysis (bottom). Each plot shows accuracy as a function of BERT layer.

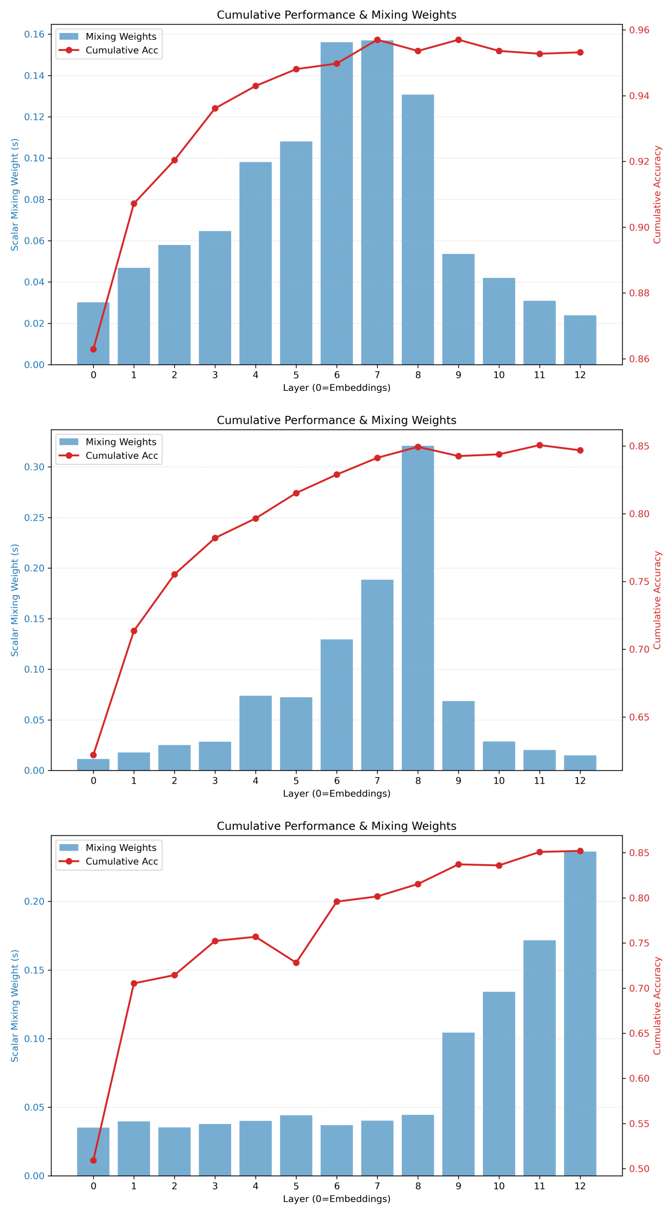

Figure 2: Scalar mixing weights (bars) and cumulative scoring (line) for POS Tagging (top), Dependency Relation Labeling (middle), and Sentiment Analysis (bottom).

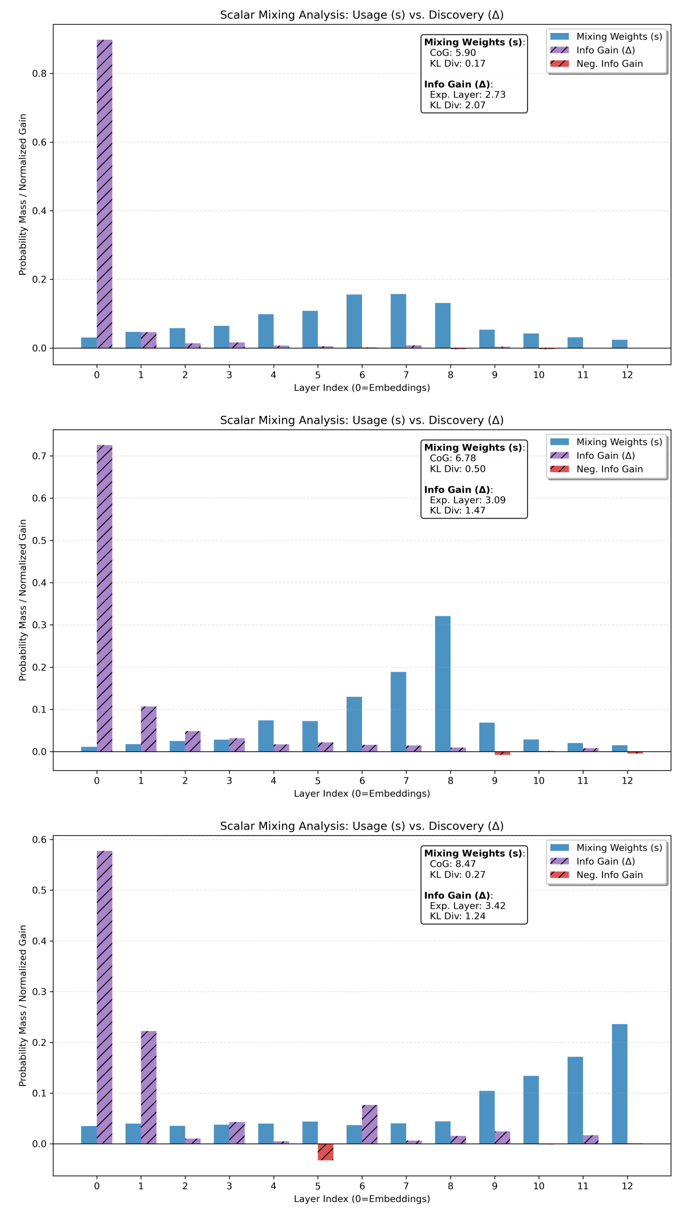

Figure 3: Scalar Mixing Analysis: Usage (s) vs. Discovery (Δ) for POS Tagging (top), Dependency Relation Labeling (middle), and Sentiment Analysis (bottom).

Figure 1 presents the layer-wise probing performance for the three tasks using logistic regression classifiers trained independently on each layer's representations. We see that:

- **POS Tagging:** Performance peaks around layer 6, indicating that morphological information is most salient in the early-to-middle layers.

- **Dependency Relation Labeling:** Accuracy reaches its maximum around layer 8, suggesting that syntactic structure is primarily encoded in the middle layers.

- **Sentiment Analysis:** The highest performance occurs at layer 11, indicating that semantic information is concentrated in the upper layers.

4.2 Scalar Mixing Weights and Cumulative Scoring

Figure 2 shows the accuracies for cumulative scoring (red line) by increasingly adding new layers, along with the contribution of each layer extracted from the learned scalar mixing weights (bars) on the full model.

Again, we observe how the learned mixing coefficients $s_l = \frac{\exp(\alpha_l)}{\sum_k \exp(\alpha_k)}$ align with our previous findings.

4.3 Usage vs Discovery

Figure 3 shows side by side the layer-wise mixing weights (usage, s) and the differential gains from cumulative scoring (discovery, Δ) for each task. Note the spike at layer 0 (embeddings) in the delta plots, which we exclude when computing the expected layer for discovery. This happens because the static embeddings already contain a significant amount of information for these tasks, but obviously do not reflect contextualized knowledge learned by BERT.

4.4 Center of gravity and Expected Layers

Table 1 agrees with our qualitative observations from the plots. We see a clear progression from lower to higher layers as we move from morphological to syntactic to semantic tasks.

Additionally, for all tasks, the Center of Gravity (CoG) is higher than the Expected Layer from differential scores, indicating that information is often discovered earlier but refined and utilized in later layers.

Lastly, the KL Divergence values suggest that syntactic information (Dependency task) is the most localized, while morphological (POS) and semantic (Sentiment) knowledge is more diffusely spread across layers.

Task	**CoG**	**Exp. Layer**	**KL Div. (Mix)**
POS	5.90	2.73	0.17
Dependency	6.78	3.09	0.50
Sentiment	8.47	3.42	0.27

Table 1: Summary statistics for each probing task: Center of Gravity (CoG) from mixing weights, Expected Layer from differential scores, and KL Divergence measuring localization.

5 Conclusion and Future Work

We have conducted a systematic layer-wise probing analysis of BERT using three tasks spanning the linguistic hierarchy: part-of-speech tagging, dependency relation labeling, and sentiment analysis. Our findings provide strong empirical evidence that BERT learns a hierarchical organization of linguistic knowledge, with morphology encoded in early layers, syntax in middle layers, and semantics in upper layers. This stratification emerges naturally from pre-training without explicit supervision.

Future work could extend this analysis to other BERT variants (e.g., RoBERTa, ALBERT) and larger models (e.g., BERT-large), but also explore additional linguistic tasks such as coreference resolution and semantic role labeling. Investigating how fine-tuning on downstream tasks affects this hierarchical structure would also be valuable.

References

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